

IOAI Machine Learning Task Recognition

Teacher Key | Suggested answers and discussion points

Use this key as a reasoning guide:

Accept equivalent wording when the student correctly identifies the model input, target/output, source of supervision, task structure, and a defensible simple baseline.

Scenario 1: Missed-Connection Risk

Input:

Passenger/itinerary features: inbound arrival time, connection time, terminal distance, airport congestion, weather, gate closing time, etc.

Output:

Probability or decision: will this passenger miss the connection / need assistance?

Label evidence:

Yes. Historical itineraries contain the actual boarding outcome for the connecting flight.

Task:

Binary classification.

Baseline:

Logistic Regression.

Metric:

F1 or PR-AUC if missed connections are relatively rare. Also inspect precision/recall depending on the operational cost of false alarms versus missed cases.

Reasoning:

The required output is a discrete yes/no outcome and historical outcomes are available.

Scenario 2: Lunch-Hour Demand

Input:

Past neighborhood order counts plus time, weather, holidays, promotions, restaurant availability, and recent demand.

Output:

Expected number of orders for each future 30-minute interval.

Label evidence:

Yes. Historical order counts provide the target values for earlier time periods.

Task:

Regression; more specifically, time-series forecasting because time order matters.

Baseline:

Seasonal-naive forecast or a simple Linear Regression using calendar/weather features.

Metric:

MAE is easy to interpret; RMSE if large misses should be penalized more strongly.

Reasoning:

The output is a numeric quantity and it is predicted for future time intervals.

Teacher Key - Continued

Scenario 3: Homepage Article Ordering

Input:

Reader history, candidate-article features, context, and historical impression/engagement records.

Output:

An ordered list of candidate articles; top 10 shown first.

Label evidence:

Historical behavior provides preference signals such as clicks, reads, dwell time, or downstream engagement. These are implicit feedback signals rather than a single clean class label.

Task:

Recommendation / ranking.

Baseline:

Popularity ranking, recency ranking, or a simple weighted score before more advanced personalized models.

Metric:

Precision@K, Recall@K, NDCG@K, or MAP depending on course level.

Reasoning:

The model's primary output is an ordering of items for a user, not a single category.

Scenario 4: Store Behavior Profiles

Input:

Monthly operational feature vectors for stores.

Output:

A small number of groups/profiles containing stores with similar behavior.

Label evidence:

No predefined target profile is described; existing geographic regions are explicitly not the desired grouping.

Task:

Clustering.

Baseline:

K-Means after appropriate scaling.

Metric:

Silhouette score plus qualitative/business interpretability. Inertia can help compare K within the same representation but should not be treated as a universal quality score.

Reasoning:

The goal is to discover groups from similarity rather than predict an existing target.

Teacher Key - Continued

Scenario 5: Unexpected Machine Behavior

Input:

Multivariate machine sensor measurements over time.

Output:

A score or flag indicating unusually different operating patterns.

Label evidence:

Only partial/weak supervision is implied: some runs led to investigations, but the goal includes previously unseen patterns. Treat the core task as anomaly detection rather than ordinary supervised classification.

Task:

Anomaly detection.

Baseline:

Isolation Forest (after sensible aggregation/scaling where needed).

Metric:

If reviewed incidents exist, use precision/recall on flagged events; otherwise combine expert review, false-alert rate, and detection rate on known incidents. Evaluation is harder without complete ground truth.

Reasoning:

The goal is to find rare deviations from ordinary operation, including novel patterns.

Common Misdiagnoses to Listen For

- "Predict" = regression. Wrong: inspect the output structure.
- "No obvious class names" = unsupervised. Wrong: historical outcomes may still provide supervision.
- "Unusual" = always anomaly detection. Wrong: with complete labeled outcomes, it may be classification.
- Recommendation = classification. Usually the core output is ranking.
- Choosing XGBoost / a neural network before identifying the task. Model choice comes after problem structure.